

ATHENS, GA, USA

WMO: 723110

Lat: 33.948N

Lon: 83.328W

Elev: 785

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.4	26.3	4.9	7.3	31.9	9.0	9.0	32.8	20.9	41.7	19.0	41.8	6.8	280	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	95.4	74.6	93.2	74.2	90.9	73.9	78.1	88.9	77.1	87.5	76.1	86.0	6.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	136.0	82.2	74.2	131.6	81.1	73.3	127.6	80.0	41.9	88.8	40.9	87.7	40.0	86.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.7	12.9		15.4	98.6	4.4	3.7	12.2	101.3	9.6	103.4	7.2	105.5	4.0	108.1
				13.3	80.1	3.9	2.0	10.5	81.6	8.2	82.7	6.0	83.9	3.2	85.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	43.9	47.7	54.6	62.3	70.2	77.3	80.1	79.3	73.6	63.3	53.1	46.5	(6)
(7)		DBStd	14.60	9.46	8.76	8.67	7.08	6.04	4.46	3.49	3.76	5.37	7.28	7.82	8.54	(7)
(8)		HDD50	667	240	136	55	5	0	0	0	0	4	55	172	(8)	
(9)		HDD65	2698	655	487	334	132	26	1	0	7	119	364	573	(9)	
(10)		CDD50	5315	50	71	198	373	627	818	934	908	709	415	147	65	(10)
(11)		CDD65	1869	0	2	12	50	188	369	469	443	265	64	6	1	(11)
(12)		CDH74	16888	1	9	124	541	1677	3432	4512	4021	2047	482	39	3	(12)
(13)		CDH80	7093	0	0	15	106	553	1547	2137	1843	787	103	2	0	(13)
(14)	Wind	WSAvg	5.6	6.8	6.8	6.9	6.1	5.2	4.7	4.4	4.3	5.0	5.1	5.4	6.1	(14)
(15)	Precipitation	PrecAvg	48.7	4.6	4.4	5.3	3.7	4.1	3.9	4.8	3.5	3.6	3.1	3.5	3.9	(15)
(16)		PrecMax	71.4	9.4	9.3	10.9	10.9	9.7	12.2	10.6	6.3	10.3	7.8	8.3	8.4	(16)
(17)		PrecMin	32.4	0.6	0.7	1.1	0.7	0.4	0.9	1.5	0.2	0.6	0.0	0.6	0.8	(17)
(18)		PrecStd	8.6	1.9	2.1	2.6	2.4	2.3	2.4	2.5	1.7	2.2	2.2	1.7	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	74.9	82.1	86.2	91.9	97.0	98.0	98.9	93.9	87.5	78.8	73.3	(19)
(20)		MCWB	59.5	61.5	63.1	67.0	70.3	73.7	75.0	75.1	72.4	70.8	64.7	61.8	(20)	
(21)		2%	DB	66.7	70.0	77.5	83.1	88.7	93.9	95.1	95.1	90.8	82.4	74.1	68.7	(21)
(22)		MCWB	58.1	57.8	61.3	65.4	70.1	73.6	75.3	74.9	72.0	67.9	60.9	61.6	(22)	
(23)		5%	DB	63.1	66.1	73.8	80.4	85.9	91.4	92.9	92.0	87.8	79.5	70.6	64.6	(23)
(24)		MCWB	56.8	56.1	59.0	64.0	69.7	73.4	75.3	74.7	71.8	65.9	60.0	58.6	(24)	
(25)		10%	DB	59.3	62.5	69.8	77.0	83.0	88.6	90.4	89.2	84.8	75.9	66.8	60.9	(25)
(26)		MCWB	53.3	54.4	57.2	61.8	68.3	72.7	74.9	74.3	70.9	64.5	58.3	55.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	65.3	66.6	70.9	75.3	78.2	79.5	76.5	74.2	69.2	67.1	(27)	
(28)		MCDWB	67.9	69.9	75.4	80.9	83.9	89.3	91.1	90.4	86.0	79.9	72.2	68.9	(28)	
(29)		2%	WB	61.4	62.4	64.1	68.5	73.4	76.8	78.2	78.1	75.2	71.5	65.9	63.5	(29)
(30)		MCDWB	65.0	67.1	72.8	78.0	82.9	87.9	89.3	88.5	84.4	78.6	69.8	66.7	(30)	
(31)		5%	WB	58.2	59.4	62.1	66.7	71.9	75.5	77.2	77.1	74.1	69.5	63.3	60.2	(31)
(32)		MCDWB	61.9	63.7	69.6	75.6	81.3	85.9	87.9	86.9	82.4	74.8	67.7	64.0	(32)	
(33)		10%	WB	54.4	56.1	60.0	64.8	70.4	74.5	76.2	75.9	72.9	67.6	60.2	56.1	(33)
(34)		MCDWB	57.8	60.4	66.6	73.3	79.2	84.4	86.3	85.0	80.5	73.5	65.3	59.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	19.4	20.1	22.1	22.9	21.8	20.4	19.7	19.2	19.6	21.3	21.4	18.7	(35)	
(36)		5% DB	MCDBR	22.4	24.2	26.7	26.5	24.8	24.3	23.1	23.5	23.6	24.7	25.2	21.9	(36)
(37)			MCWBR	15.1	14.7	13.4	11.6	9.4	7.8	7.0	7.3	8.2	11.3	13.9	14.5	(37)
(38)			5% WB	MCDWB	18.4	19.1	21.6	21.5	20.3	19.9	19.9	19.4	18.7	18.3	18.6	17.9
(39)		MCWBR		14.5	14.1	13.0	11.3	9.1	7.8	7.4	7.2	7.7	10.1	13.3	14.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.319	0.355	0.390	0.451	0.480	0.520	0.518	0.422	0.383	0.341	0.310	(40)	
(41)		taud	2.519	2.495	2.422	2.312	2.180	2.146	2.054	2.046	2.298	2.388	2.455	2.551	(41)	
(42)		Ebn at Noon	286	296	292	285	267	258	247	245	266	269	272	281	(42)	
(43)		Edh at Noon	26	30	35	41	47	49	53	53	39	33	28	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	866	1053	1395	1741	1926	1993	1892	1738	1508	1270	958	738	(44)	
(45)		RadStd	49	127	99	116	151	151	129	89	145	148	94	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+279	+185	(46)
(47)	Regional (42 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+202	(47)

Nomenclature: See separate page